AEGIS: Closed-Form Adversarial Diagnostics over the GNN Vulnerability Spectrum

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***Abstract*—Graph neural networks deployed in safety-critical settings (drug interaction, fraud detection, power-grid contingency screening) are vulnerable to adversarial graph perturbations, yet no prior method covers the full adversarial vulnerability spectrum a practitioner needs from one closed-form computation: a globally maximally sensitive perturbation direction, per-edge equilibrium-sensitivity rankings, and per-node first-order sensitivity radii (first-order diagnostics, complementary to sound certificates such as AGNNCert). AEGIS produces all three diagnostics across this spectrum from one object, the *constrained sensitivity matrix* S_c, computed matrix-free via a Neumann-series resolvent and randomized SVD that scale to N=7,650 on a single GPU. The analytical construction applies to any GNN with continuous edge-weight-modulated message passing (GCN, SAGE, GIN, APPNP, IGNN, edge-weighted GAT†); predictive transfer is architecture-dependent, with deeper-than-2-layer models transferring most reliably (29/33 architecture–dataset cells positive, sign-test $p<10^{-5}$). For the spectral-norm-constrained IGNN subclass ($\sim6\%$ Cora accuracy cost) we additionally derive a critical budget $\varepsilon_{\text{crit}}=(1-\kappa)/\|W\|_2$ and a worst-case three-regime characterisation, empirically anchored by a $\kappa_{\max}$ sweep (section V-D). Tightness lies in $[1.00, 1.39]$ over $\varepsilon\in[0.01, 0.20]$ across 7 architectures and 9 datasets; the SVD direction reaches 0.72–0.92$\times$ of an IFT-gradient PGD baseline; non-adaptive top-v_{ij} edge masking reduces SVD damage by $42\pm8\%$ vs. $11\pm6\%$ random ($p<0.002$). On IEEE power-flow benchmarks (proof-of-concept, not operator-grade), S_c correlates with N-1 contingency rankings on case14–118 ($\tau=+0.37$ to $+0.62$, P@10 $=0.66$–0.81); code will be released under tiered access.**

***Index Terms*—Graph neural networks, adversarial robustness, vulnerability spectrum, constrained sensitivity, deep equilibrium models, implicit function theorem**

I. INTRODUCTION

Graph neural networks are now deployed in domains where structural errors carry tangible consequences: fraudulent accounts evade detection when an adversary inserts edges into a financial graph [1], drug-interaction models misfire under perturbed molecular graphs [2], and power grids can miss critical contingencies when their graph representations are structurally fragile [3]. A practitioner deploying a GNN faces a question current tools leave unanswered: which edges, if perturbed, would cause predictions to fail, and by how much?

Structural attacks [1], [4], [5] give per-edge surrogate gradients but no continuous direction or per-node budgets; smoothing [6], [7], [8], [9] and IBP-style certifiers [10], [11] give per-node certificates but no edge rankings or direction; robust-architecture defenses [12], [13], [14] harden models without surfacing a vulnerability spectrum; sober-look critiques [15], [16] flag evaluation pitfalls. AEGIS (Adversarial Evaluation of Graph Integrity via Sensitivity) covers the full vulnerability spectrum with three closed-form diagnostics from one analytical object, S_c (section IV); the formal regime guarantees of theorem 1 apply to contractive implicit GNNs, and S_c extends as a computational tool to any GNN with continuous edge-weight-modulated message passing (section V-F).

Contributions. (1) Constrained sensitivity operator. S_c and a matrix-free pipeline yielding the SVD-optimal direction, per-edge rankings, and per-node radii from one closed-form computation, scaling to N=7,650 (matrix-free/dense σ_1 agree to 0.03% at N=200). **(2) Phase-transition theory.** A worst-case three-regime characterisation with critical budget $\varepsilon_{\text{crit}}=(1-\kappa)/\|W\|_2$ for the spectral-norm-constrained IGNN subclass ($\sim6\%$ accuracy cost; theorem 1). **(3) Empirical evaluation.** Over 9 datasets, 7 architectures, 4 domains (330 runs), with four-quadrant attack comparison, head-to-head vs. GR-BCD/PR-BCD [5]/AGNNCert/LODF (table IV), continuous-to-discrete transfer (fig. 7), and a power-grid case study (section VI).

II. PRELIMINARIES AND THREAT MODEL

Let $G=(V,E,A)$ be an undirected graph with $N=|V|$ nodes and adjacency $A\in\mathbb{R}^{N\times N}$, node features $X\in\mathbb{R}^{N\times d_{\text{in}}}$, and symmetric normalized adjacency $\hat{A}=D^{-1/2}(A+I)D^{-1/2}$ (self-loops included). Throughout, ϕ is the activation, $\sigma_i(\cdot)$ the i-th singular value, and $\text{vec}(\cdot)$ the column-major vectorization. The sensitivity objects introduced below are: the unconstrained $S\in\mathbb{R}^{Nd\times N^2}$, constrained $S_c\in\mathbb{R}^{Nd\times|E|}$, unrolled $S_K\in\mathbb{R}^{Nd\times N^2}$ (theorem 7), and per-node block S_v; the contractivity $\kappa=\|J_z\|_2$, spectral radius $\rho(J_z)$, pseudospectral index $\eta\geq1$, critical budget $\varepsilon_{\text{crit}}$, and per-node radius r_v.

IGNN and IFT. A DEQ-GNN [17], [18] iterates a weight-tied F_θ to a fixed point; the IGNN instantiation [19] uses

$$F(Z,A)=\phi(\hat{A}ZW^\top+X_{\text{proj}}),\tag{1}$$

with $W\in\mathbb{R}^{d\times d}$ spectral-normalized [20] so $\kappa\leq\left\|\hat{A}\right\|_2\|W\|_2<1$, ReLU ϕ, and a learned projection X_{proj}. Training uses implicit differentiation through the fixed point [21], [22] with optional Jacobian regularization [23], and a linear head $f(z^*)=W_{\text{cls}}z^*+b$ produces class predictions. At equilibrium, $G(z^*,A)=z^*-F(z^*,A)=0$ defines z^* as an implicit function of A, with IFT sensitivity

$$\frac{\partial z^*}{\partial p}=(I-J_z)^{-1}\frac{\partial F}{\partial p}\bigg|_{z^*},\tag{2}$$

where $\left\|(I-J_z)^{-1}\right\|_2\leq1/(1-\kappa)$ via the convergent Neumann series, and the pseudospectral index $\eta=\left\|(I-J_z)^{-1}\right\|_2(1-\rho)\geq1$ [24] measures the gap to the spectral-radius bound (with

η=1 for normal J_z). The IFT has been used for hyperparameter optimization [25] and implicit differentiation [21], but not for adversarial structural analysis.

Threat model. A white-box attacker with full access to the trained IGNN perturbs the normalized adjacency

$$\hat{A}' = \hat{A} + \delta\hat{A}, \quad \left\|\delta\hat{A}\right\|_F \leq \varepsilon, \tag{3}$$

with $\delta\hat{A}$ symmetric ($\delta\hat{A}$=$\delta\hat{A}^\top$, preserving the undirected graph), edge-only ($[\delta\hat{A}]_{ij}$=0 for $(i,j) \notin E$, so no new edges), and continuous in $\mathbb{R}$. AEGIS targets edge-deletion / weight-perturbation attacks; insertion attacks (e.g., Nettack [1], Mettack [4]) require enlarging the basis from E to a candidate set $\bar{E} \supseteq E$ and lie outside our threat model. Discrete deletions are connected to the continuous model through theorem 6. Unlike input-feature perturbation [26], [27], structural perturbation changes the operator itself: z^* depends on A through both $J_z = \operatorname{diag}(\phi')(\hat{A} \otimes W)$ and $J_A = \partial F/\partial \operatorname{vec}(A)$, which is what makes structural sensitivity qualitatively different and motivates the construction of S_c in section III.

III. Adversarial Equilibrium Theory

Building on the threat model of section II, we now derive the analytical machinery behind the three diagnostics promised in section I: per-edge rankings, the SVD-optimal direction, and per-node radii. The main result (theorem 1) characterizes how vulnerability evolves with perturbation magnitude, and the constrained sensitivity matrix S_c provides the unified object from which the three diagnostics follow.

Theorem 1 (Vulnerability Characterization for Contractive Implicit GNNs)**.** *Let $F_\theta(Z, \hat{A}) = \phi(\hat{A}ZW^\top + X_{\text{proj}})$ be an IGNN-class operator satisfying:*

(A1) *ϕ is ReLU or any 1-Lipschitz activation; nonsmoothness at zero is handled via the conservative IFT [28] (proof below).*
(A2) *$\|W\|_2 \leq c$ via spectral normalization.*
(A3) *$\|J_z\|_2 \leq \kappa < 1$ verified post-training (κ=0.14–0.59 across datasets; section V-E). (A2) does not imply (A3) automatically; we report κ alongside $\varepsilon_{\text{crit}}$ in our experiments so practitioners can audit. If (A3) fails,* AEGIS *still returns S_c rankings and SVD-optimal attacks but flags that the formal guarantees below are void.*

Define the critical budget

$$\varepsilon_{\text{crit}} = \frac{1-\kappa}{\|W\|_2}. \tag{4}$$

Then structural perturbations $\delta\hat{A}$ with $\left\|\delta\hat{A}\right\|_F = \varepsilon$ exhibit three regimes:

(a) Subcritical *($\varepsilon < \varepsilon_{\text{crit}}$): The perturbed operator remains contractive with a unique fixed point satisfying*

$$\|\Delta z^*\|_F \leq \sigma_1(S) \cdot \varepsilon + O(\varepsilon^2), \tag{5}$$

where $S = (I - J_z)^{-1} J_A$ is the structural sensitivity matrix and $\sigma_1(S) \leq \|J_A\|_{\text{op}}/(1-\kappa)$.

(b) Critical *($\varepsilon \to \varepsilon_{\text{crit}}$): For worst-case directions that maximally increase $\left\|\hat{A}+\delta\hat{A}\right\|_2$ (aligned with the top singular vector of $\hat{A}$), $\left\|(I-J_z')^{-1}\right\|_2$ diverges as $\Omega(1/(\varepsilon_{\text{crit}}-\varepsilon))$ from the Neumann lower bound $1/(1-\|J_z'\|_2)$, which holds without normality assumptions on J_z'. Generic directions diverge more slowly because $\|J_z'\|_2$ need not approach 1.*

(c) Supercritical *($\varepsilon \geq \varepsilon_{\text{crit}}$): The sufficient contraction certificate becomes void ($\|J_z'\|_2$ may exceed 1). The perturbed operator may still be contractive along low-sensitivity directions, but Banach no longer guarantees uniqueness and the part-(a) first-order guarantees lapse; we do not claim divergence as a generic post-threshold behavior, only that our certificate fails (empirical $\kappa_{\max}$ sweep: section V-D).*

Proof. **(a).** Under (A1), F_θ is piecewise-affine on each ReLU linear region; the activation pattern at z^* is generically stable for sufficiently small $\left\|\delta\hat{A}\right\|$ (the boundary set $\{\hat{A} : z^*(\hat{A})$ lies on a region face$\}$ has Lebesgue measure zero), so the conservative-gradient calculus of Bolte–Pauwels [28] yields a conservative IFT on each region. With $J_z = \operatorname{diag}(\phi')(\hat{A} \otimes W)$ and $\kappa \leq \left\|\hat{A}\right\|_2 \|W\|_2$, applying IFT to $G(z,\hat{A})$=$z-F(z,\hat{A})$=0 at $(z^*, \hat{A})$ gives

$$\Delta z^* = (I - J_z)^{-1} J_A \cdot \operatorname{vec}(\delta\hat{A}) + O(\left\|\delta\hat{A}\right\|^2), \tag{6}$$

with $J_A = \partial F/\partial \operatorname{vec}(A)$ and Neumann $\left\|(I - J_z)^{-1}\right\|_2 \leq 1/(1-\kappa)$ yielding (5). For contractivity preservation, $\|J_z'\|_2 \leq (\left\|\hat{A}\right\|_2 + \varepsilon)\|W\|_2$, so $\varepsilon < \varepsilon_{\text{crit}}$ guarantees $\|J_z'\|_2 < 1$.

(b)–(c). Along the worst-case direction $\|J_z'\|_2 \to \left\|\hat{A}\right\|_2 \|W\|_2 + \varepsilon \|W\|_2$, so $1-\|J_z'\|_2 \geq \|W\|_2 (\varepsilon_{\text{crit}}-\varepsilon)$; the Neumann lower bound $\left\|(I-J_z')^{-1}\right\|_2 \geq 1/(1-\|J_z'\|_2)$ requires no normality assumption and gives the $\Omega(1/(\varepsilon_{\text{crit}}-\varepsilon))$ rate. For $\varepsilon \geq \varepsilon_{\text{crit}}$ Banach fails; the iteration may converge elsewhere, oscillate, or diverge. □

$\varepsilon_{\text{crit}}$ is sufficient, not necessary, and the unconstrained $\sqrt{r}$ slack is 7–14× on 50-node subgraphs; S_c closes this in practice (tightness ≈1.00 at ε=0.01; section V-A). The Neumann series requires the operator-norm κ rather than $\rho(J_z)$ [29]; the gap is the pseudospectral index η, bounded by the weight matrix alone:

Observation 2 (Graph-Independent Nonnormality Bound)**.** *For $J_z = \operatorname{diag}(\phi')(\hat{A} \otimes W)$ with symmetric $\hat{A}$: if all activations are positive, $\eta \leq \kappa(V_W)$ where W=$V_W D_W V_W^{-1}$ and $\kappa(V_W)$=$\|V_W\|_2 \left\|V_W^{-1}\right\|_2$, independent of $\hat{A}$. For general ReLU patterns $\eta \leq \kappa(V_{J_z})$, which empirically stays near $\kappa(V_W)$ (η=1.19–2.47; section V-E).*

Proof. When $\phi' \equiv 1$, J_z=$\hat{A} \otimes W$ has Kronecker eigenvalues $\lambda_i(\hat{A})\lambda_j(W)$ [29]; symmetric $\hat{A}$ gives $\kappa(U_{\hat{A}})$=1, so the joint eigenvector matrix has condition number $\kappa(V_W)$, and the diagonalisable bound $\left\|(I-M)^{-1}\right\|_2 \leq \kappa(V)/(1-\rho(M))$ yields $\eta \leq \kappa(V_W)$. Graph structure does not amplify nonnormality; the moderate empirical η traces to the spectral-norm regularisation of W.

Proposition 3 (Maximally Sensitive First-Order Structural Perturbation)**.** *The perturbation δA^* maximizing $\|\Delta z^*\|$ to first order subject to $\|\delta A\|_F \le \varepsilon$ is $\delta A^* = \varepsilon \cdot \mathrm{reshape}(v_1, N \times N)$, where v_1 is the leading right singular vector of S. The maximum first-order equilibrium shift is $\varepsilon \cdot \sigma_1(S)$.*

By the variational characterisation of σ_1, the direction is optimal for hidden-state damage and empirically aligned with classification damage (section V-B). The *constrained sensitivity matrix* $S_c \in \mathbb{R}^{Nd \times |E|}$ has columns $[S_c]_{:,k} = S_{:,iN+j} + S_{:,jN+i}$ paired with edge basis $b_k{=}(e_i e_j^\top + e_j e_i^\top)/\sqrt{2}$ ($\|b_k\|_F{=}1$, so $\sigma_1(S_c)$ is consistent with $\left\|\delta\hat{A}\right\|_F{=}\varepsilon$). The $N^2 \to |E|$ projection P_c is the standard duplication-matrix reduction [30] on the edge-supported symmetric subspace; the operational contribution is the matrix-free $S_c v$ in algorithm 1. The envelope ratio is tight (≈ 1.00) at $\varepsilon{=}0.01$ and reaches ≤ 1.39 at $\varepsilon{=}0.20$, identifying the dangerous direction rather than a global safety margin.

Proposition 4 (Per-Node First-Order Sensitivity Radius)**.** *For a linear head $f(z){=}Wz{+}b$ and node v with margin $m_v{=}f_{y_v}(z_v^*){-}\max_{c\neq y_v} f_c(z_v^*) > 0$, the first-order sensitivity radius*

$$r_v = \frac{m_v}{\|W_{y_v} - W_{c^*}\|_2 \cdot \|S_v\|_2} \tag{7}$$

preserves the classification of v for any $\left\|\delta\hat{A}\right\|_F < r_v$, where S_v are the block-rows of S at node v. Under the constrained threat model the tighter radius $r_v^{(c)} = m_v/(\|W_{y_v}{-}W_{c^*}\|_2 \, \|S_{c,v}\|_2) \ge r_v$ holds, using the block-rows of S_c at node v.

Proof sketch. By Thm. 1(a), $\Delta z_v^* \approx S_v \cdot \mathrm{vec}(\delta A)$, so $|\Delta f_{y_v}| \le \|W_{y_v}{-}W_{c^*}\|_2 \, \|S_v\|_2 \, \|\delta A\|_F$; misclassification requires $|\Delta f_{y_v}| \ge m_v$, and inverting gives r_v. Intuitively, r_v is the margin divided by two amplification factors (decision-boundary sensitivity to hidden-state change, and equilibrium sensitivity to structural perturbation); high-margin, low-sensitivity nodes get large radii.

Remark 5 (Sensitivity threshold semantics)**.** *r_v is a first-order sensitivity threshold, not a probabilistic certificate: it indicates the perturbation scale at which the linearization predicts a classification change for node v, and is locally tight for small ε but can be violated at larger magnitudes where higher-order terms dominate. The two approaches are complementary: randomized smoothing [6] certifies which nodes are guaranteed safe, while* AEGIS *differentiates which edges and nodes are most sensitive and in which direction (section V-B). Empirically, breach rate is 0% at $\varepsilon{=}0.01$ and rises with ε as expected for first-order approximations (section V-B).*

A. Continuous-to-Discrete Transfer

Proposition 6 (Continuous-to-Discrete Ranking Transfer)**.** *Under (A1)–(A3), let $w_k = [\hat{A}]_{ij}$ denote the normalized weight of edge $k = (i,j) \in E$, $v_k = \|[S_c]_{:,k}\|_2$ the continuous vulnerability score, and $d_k = \|z^*(A) - z^*(A \setminus k)\|_2$ the discrete damage from full removal of edge k. Assume all single-edge removals are subcritical: $\sqrt{2}\max_k w_k < \varepsilon_{\mathrm{crit}}$.*

(a) First-order bridge. *The discrete damage satisfies*

$$d_k = w_k \cdot v_k + R_k, \quad |R_k| \le \frac{L_J}{2(1-\kappa)^2} \cdot w_k^2, \tag{8}$$

where L_J is the Lipschitz constant of the state Jacobian J_z with respect to $\mathrm{vec}(A)$.

(b) Ranking preservation. *For two edges k_1, k_2 with $v_{k_1} > v_{k_2}$ and $w_{k_1} \ge w_{k_2}$, we have $d_{k_1} > d_{k_2}$ whenever*

$$w_{k_1} < \frac{v_{k_1} - v_{k_2}}{L_J/(1-\kappa)^2}. \tag{9}$$

Under uniform weights ($w_k = w$ for all k), $v_{k_1} > v_{k_2}$ implies $d_{k_1} > d_{k_2}$ whenever $w < \min_{k_1 \neq k_2}(v_{k_1} - v_{k_2}) \cdot (1-\kappa)^2/L_J$.

Proof. **(a).** Edge removal corresponds to $[\delta\hat{A}]_{ij}{=}[\delta\hat{A}]_{ji}{=}{-}w_k$, so by the S_c construction $\|\Delta z^*\| \approx w_k v_k$. For the remainder, let $g(t){=}z^*(A{+}t\delta\hat{A}_k)$, $t{\in}[0,1]$. IGNN is piecewise affine in A for a fixed activation pattern, so $g(t)$ is piecewise linear and crosses finitely many activation boundaries; on each segment J_z is constant. Across a boundary J_z jumps by at most $\|W\|_2^2$ (the max change in $\mathrm{diag}(\phi')(\hat{A} \otimes W)$ when one activation flips), so the worst-case path Lipschitz constant $L_J \le \|W\|_2^2$. The resolvent identity $\frac{d}{dt}(I-M)^{-1}{=}(I-M)^{-1}\frac{dM}{dt}(I-M)^{-1}$ then yields $\|g''(\xi)\| \le (1{-}\kappa)^{-2} L_J w_k^2$, hence $|R_k| \le L_J w_k^2/(2(1{-}\kappa)^2)$. **(b).** From (a), $d_{k_1}{-}d_{k_2} \ge w_{k_1}v_{k_1}{-}w_{k_2}v_{k_2}{-}C(w_{k_1}^2{+}w_{k_2}^2)$ with $C{=}L_J/(2(1-\kappa)^2)$; under $v_{k_1}{>}v_{k_2}$, $w_{k_1}{\ge}w_{k_2}$, the first term is positive and (9) ensures the first-order gap dominates. ☐

Empirically $|R_k|$ is 2–10$\times$ smaller than $L_J w_k^2$, and the $29/33$ positive-transfer cells of fig. 7 confirm the piecewise structure does not erode the bridge: shallow models with near-uniform v_k (GCN-2) violate part-(b), deeper models satisfy it. The ingredients of theorem 1 are classical [29], [24]; the contribution is the constrained projection S_c that turns a vacuous bound (tightness 0.31) into a tight one (1.00; section V-A) and unifies attacks, rankings, and radii.

B. Generalization to Explicit GNNs

Observation 7 (Structural Sensitivity for K-Layer GNNs)**.** *Let $Z_K = g_K \circ \cdots \circ g_1(X, A)$ be a K-layer GNN where each layer g_l is differentiable with respect to both its input and A. Define the* unrolled sensitivity matrix

$$S_K = \frac{\partial \mathrm{vec}(Z_K)}{\partial \mathrm{vec}(A)} = \sum_{l=1}^{K} \left(\prod_{k=l+1}^{K} J_z^{(k)} \right) J_A^{(l)}, \tag{10}$$

where $J_z^{(k)} = \partial g_k / \partial Z_{k-1}$ and $J_A^{(l)} = \partial g_l / \partial \mathrm{vec}(A)$. Then:

(a) *The first-order shift bound $\|\Delta Z_K\|_F \le \sigma_1(S_K) \cdot \|\delta A\|_F + O(\|\delta A\|^2)$ holds, with*

$$\sigma_1(S_K) \le \sum_{l=1}^{K} \left(\prod_{k=l+1}^{K} \left\|J_z^{(k)}\right\|_2 \right) \left\|J_A^{(l)}\right\|_2. \tag{11}$$

(b) *The constrained construction S_c and per-node radius r_v (theorem 4) apply identically with S_K in place of S.*

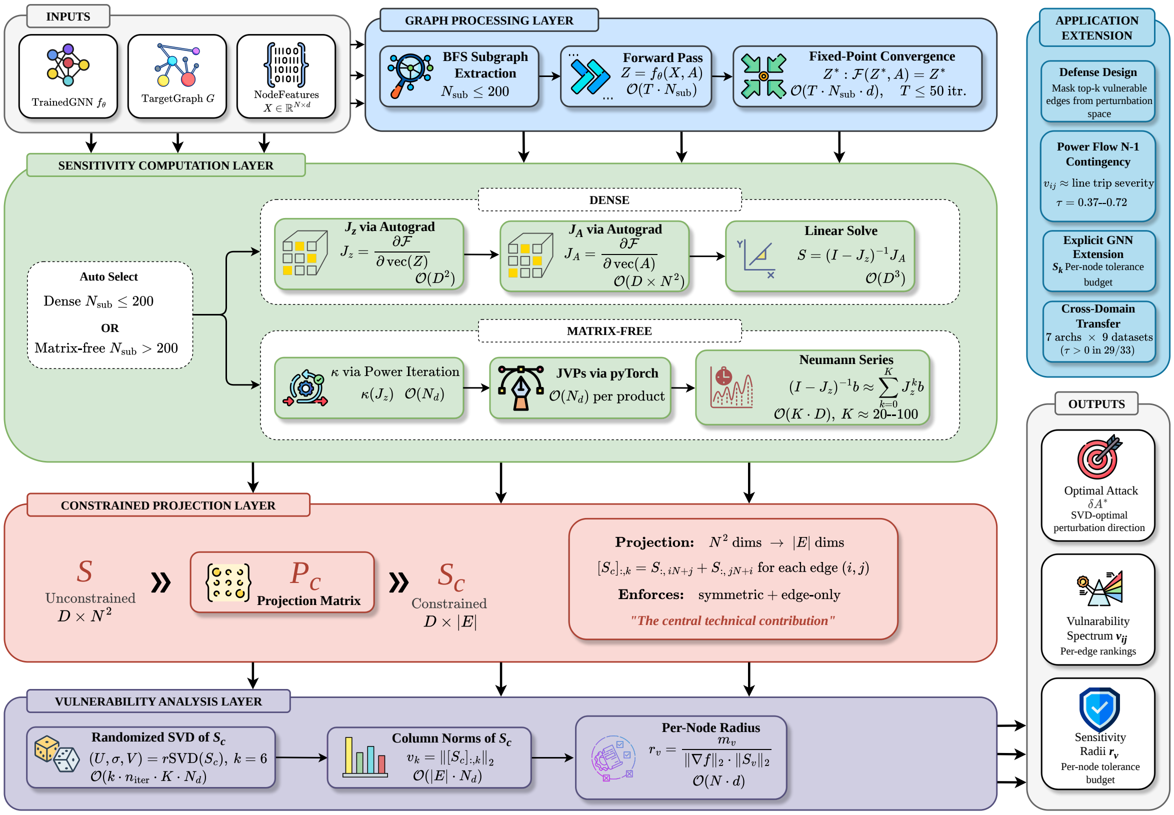


Fig. 1: The four-stage AEGIS pipeline: (1) graph processing (BFS subgraph or full-graph forward / fixed-point pass); (2) sensitivity computation (dense Jacobian for $N \leq 200$, matrix-free Neumann/JVP otherwise); (3) constrained projection $N^2 \to |E|$ via P_c; (4) vulnerability analysis (randomised SVD, S_c column norms, per-node radii). The matrix-free path queries S_c via JVPs/VJPs without materialising it, scaling to N=7,650.

For weight-tied models, $\sigma_1(S_K) \leq \|J_A\|_2 \,(1-\kappa^K)/(1-\kappa)$ converges to the IGNN bound as $K\to\infty$. Explicit GNNs inherit the practical tools but lack $\varepsilon_{\text{crit}}$ and the regime characterisation, which depend on (A3); a Jacobian product $\prod_k \left\|J_z^{(k)}\right\|_2 > 10$ diagnoses unreliable first-order predictions, and the empirical tightness 0.99–1.02 (section V-F) validates the approximation regardless of the a priori bound.

IV. AEGIS CONSTRUCTION

The pipeline turns section III into a practical tool through matrix-free computation, sidestepping both the materialisation of $S \in \mathbb{R}^{Nd\times N^2}$ and the formation of the resolvent. One S_c computation returns the per-edge spectrum $\{v_{ij}\}$, the SVD-optimal direction $\delta\hat{A}^\star$, and per-node radii $\{r_v\}$, plus $\varepsilon_{\text{crit}}$ and the regime characterisation of theorem 1 for IGNN-class operators. A dense path ($N \leq 200$) computes an exact SVD; a matrix-free path ($N > 200$, validated to N=7,650) uses JVP/VJP queries and randomized SVD. Figure 1 illustrates the four stages; algorithm 1 summarises the matrix-free routing.

Matrix-free operator. $S_c v = (I-J_z)^{-1} J_A P_c v$ is evaluated via the truncated Neumann series $\sum_{k=0}^{K} J_z^k b$ with depth K adaptive in κ (early stop at $\left\|J_z^k b\right\| < 10^{-6}\,\|b\|$); each Neumann term and $J_A \operatorname{vec}(\delta A)$ are forward-mode JVPs in $O(Nd)$ time. P_c is applied implicitly, so neither S_c nor S is formed; the dense path ($N \leq 200$) materialises J_z, J_A and solves via `torch.linalg.solve`. The rSVD [31] (k=p=10, n_{iter}=2) returns (u_1, σ_1, v_1) for $\delta\hat{A}^\star$=$\varepsilon\cdot\text{sym}(P_c v_1)$; column norms give $\{v_{ij}\}$, and per-node radii follow theorem 4 via the node-restricted RMATVEC. The well-separated singular spectrum (gap $(\sigma_1-\sigma_2)/\sigma_1$=0.39–0.50; fig. 2) is why one closed-form SVD matches iterative attackers (section V-B).

Cost. Each $S_c v$ is $O(K\cdot Nd)$ time and $O(Nd)$ memory ($K\in[20, 50]$ for $\kappa<0.8$); removing the $O((Nd)^3)$ solve lifts the previous $N\approx 300$ ceiling (section V-D).

V. EXPERIMENTS

Setup. We evaluate AEGIS on 9 datasets across 4 domains with 10 random seeds each (PyTorch [32]). IGNN [19] uses $F(Z)$=ReLU$(\hat{A}ZW^\top+U(X))$ with $W \in \mathbb{R}^{64\times 64}$ spectral-normalised [20] and Adam (lr 0.01); ContractiveGCN-PF mirrors this with 5 bus features and 2-head readout. Subgraphs: 50-node BFS from the highest-degree node. The spectral-norm constraint costs IGNN ∼6% accuracy (77.5% vs. ∼83% on Cora), the price of the guarantees in theorem 1. Datasets: Cora, Citeseer, Pubmed [33], Amazon Photo [34], WikiCS [35], IEEE 14/30/57/118 [36], [37]. Baselines: smoothing (Gaussian

Algorithm 1 AEGIS Vulnerability Analysis (matrix-free). Assumes the equilibrium z^* is pre-computed on the analysis (sub)graph. Dense fallback ($N \leq 200$): form $S = (I - J_z)^{-1} J_A$ and run a deterministic SVD in place of lines 2–11.

Require: F, z^*, $\hat{A}$, budget ε; SVD rank k, oversampling p, power iters n_{iter}; Neumann tol. τ; edge set E; (opt.) head h, logits Y, labels y.
Ensure: $\sigma_1(S_c)$, $\delta\hat{A}^\star$, $\{v_{ij}\}$, $\{r_v\}$, $\varepsilon_{\text{crit}}$.
1: $\kappa \leftarrow \|J_z\|_2$ via power iteration; $K \leftarrow \lceil \log(1/\tau)/\log(1/\kappa) \rceil$.
2: $\text{MATVEC}(v) := \sum_{j=0}^{K} J_z^j \text{JVP}_F^{\hat{A}}(P_c v)|_{z^*}$; $\text{RMATVEC}(u) := P_c^\top \text{VJP}_F^{\hat{A}}(\sum_{j=0}^{K} (J_z^\top)^j u)|_{z^*}$.
3: Draw $\Omega \sim \mathcal{N}(0, I) \in \mathbb{R}^{|E| \times (k+p)}$; $Y \leftarrow \text{MATVEC}(\Omega)$ column-wise.
4: **for** $j = 1, \ldots, n_{\text{iter}}$ **do**
5: $\quad Y \leftarrow \text{QR}(\text{MATVEC}(\text{RMATVEC}(Y)))$
6: **end for**
7: $Q \leftarrow \text{QR}(Y)$; $B^\top \leftarrow \text{RMATVEC}(Q)$; $(\widetilde{U}, \Sigma, V^\top) \leftarrow \text{SVD}(B)$; $U \leftarrow Q\widetilde{U}$ {rSVD [31]}
8: $\sigma_1 \leftarrow \Sigma_{1,1}$; $\delta\hat{A}^\star \leftarrow \varepsilon \cdot \text{sym}(P_c V_{:,1}) / \|\text{sym}(P_c V_{:,1})\|_2$.
9: $v_{ij} \leftarrow \|\text{MATVEC}(e_{ij})\|_2$ for each $(i,j) \in E$ {per-edge spectrum}
10: **if** h, Y, y supplied **then**
11: $\quad r_v \leftarrow m_v / (\|W_{y_v} - W_{c^\star}\|_2 \|S_{c,v}\|_2)$ for each v via 10 power iters {Prop. 4}
12: **end if**
13: **if** $\|W\|_2$ extractable **then** $\varepsilon_{\text{crit}} \leftarrow (1-\kappa)/\|W\|_2$ {Thm. 1}
14: **return** $(\sigma_1, \delta\hat{A}^\star, \{v_{ij}\}, \{r_v\}, \varepsilon_{\text{crit}})$.

σ=0.01, 200 MC, Clopper–Pearson α=0.001), Mettack with Meta-Self GCN surrogate, LODF as $\text{PTDF}_{ij}/(1-\text{PTDF}_{jj})$ from the DC B-matrix [38, Ch. 11]. Code and checkpoints will be released under tiered access (attack-generation behind institutional-affiliation review).

A. *Cross-Domain Adversarial Analysis*

Unless otherwise stated, IGNN experiments in section V-A–section V-E use 50-node BFS ego-subgraphs centred on the highest-degree node and report *local vulnerability*; on Cora (N=2,708), 50-node rankings correlate weakly with full-graph rankings (τ=0.16; section V-E), so graph-level assessment uses the matrix-free pipeline (section V-D). All bounds use the operator-norm $\kappa = \|J_z\|_2$; on our suite, the κ-based $\varepsilon_{\text{crit}}$ runs up to 53% tighter than the spectral-radius counterpart (worst on Citeseer; pseudospectral index η=1.19–2.47). Table I verifies (A3) on all benchmarks ($\kappa < 1$) and reports the per-dataset critical budget $\varepsilon_{\text{crit}}$ and attack advantage.

Table II reports tightness degradation across the five datasets and four ε values.

The envelope ratio is tight (1.00–1.02) at ε=0.01 and grows to 1.05–1.39 at ε=0.20; the bound is reliable only at small ε and under-predicts by up to 39% at the largest budget. The ratio is measured on equilibrium shift $\|\Delta z^*\|_F$, distinct from

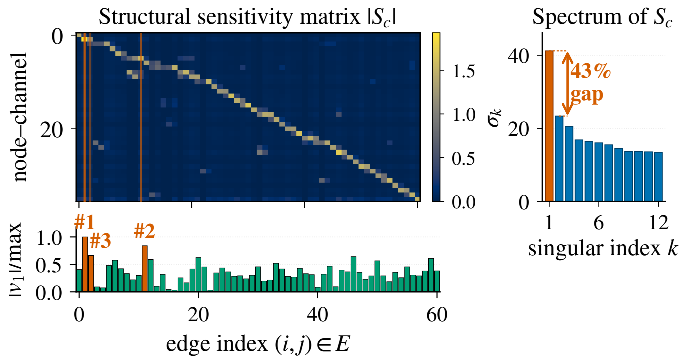


Fig. 2: $|S_c|$ on a 50-node Cora ego-graph (IGNN, N=50, d=16, $|E|$=61, seed 0). Vermilion lines: top-3 edges by $|v_1|$ (SVD-optimal direction, bar strip). Sidebar: singular spectrum, σ_1=41.2, 43% gap to σ_2 (cross-benchmark $(\sigma_1-\sigma_2)/\sigma_1 \in [0.39, 0.50]$; section V-B).

TABLE I: Per-dataset IGNN characterisation (10 seeds). $\kappa = \|J_z\|_2$ verifies (A3); $\varepsilon_{\text{crit}} = (1-\kappa)/\|W\|_2$ is the theoretical critical budget. AtkAdv: AEGIS damage / random damage. Cov%: fraction of subgraph nodes with positive first-order radius r_v.

Dataset	Test Acc	κ	$\varepsilon_{\text{crit}}$	AtkAdv	Cov%
Cora	$77.5_{\pm 1.7}$	$.33_{\pm .14}$	$.66_{\pm .15}$	$3.6_{\pm .6}$	$81_{\pm 9}$
Citeseer	$66.0_{\pm 0.7}$	$.59_{\pm .02}$	$.41_{\pm .02}$	$4.1_{\pm .5}$	$83_{\pm 4}$
Pubmed	$78.9_{\pm 0.7}$	$.41_{\pm .11}$	$.59_{\pm .11}$	$3.2_{\pm .4}$	$74_{\pm 23}$
Amazon	$94.8_{\pm 0.3}$	$.14_{\pm .02}$	$.86_{\pm .02}$	$3.8_{\pm .2}$	$92_{\pm 4}$
WikiCS	$77.9_{\pm 0.4}$	$.34_{\pm .04}$	$.66_{\pm .04}$	$3.8_{\pm .4}$	$70_{\pm 3}$

the prediction-flip (breach) rate of fig. 4 (flips stay $< 2\%$ at $\varepsilon \leq 0.10$, $< 6\%$ at ε=0.20).

Heuristic and surrogate baselines. AEGIS beats degree-proportional, edge-betweenness, and spectral rankings at ε=0.01 (+6–148% AtkAdv, Wilcoxon $p<0.001$) and inflicts 3–10× more damage than Mettack [4] (149/150 wins, one-sided sign-test $p<10^{-43}$); GR-BCD/PR-BCD [5] are in table IV. The distinctive value appears under discrete removal (fig. 3).

B. *Four-Quadrant Attack Comparison*

Radius vs. smoothing. Against randomized smoothing [6] ($\sigma \in \{0.01, 0.05, 0.10, 0.15\}$, 1000 MC samples, 10 seeds), base smoothing radii are constant per node ($r = \sigma \cdot \Phi^{-1}(p_{\text{lower}})$, e.g. 0.123 vs. AEGIS's 0.046 at σ=0.05) and offer no per-edge differentiation; localized smoothing [8] gives per-node deterministic-radius probabilistic certificates at 10^3–10^4 MC samples per node but still lacks an edge structure or attack direction. AEGIS radii are structurally informative ($r_v \approx 0.09$ in dense regions, 0.01 at boundaries), placing the three methods at complementary points on the certificate-vs-diagnostic frontier.

Table III evaluates AEGIS against baselines spanning the four quadrants of (gradient-based, gradient-free) × (equilibrium-shift, classification-loss). Three observations follow: 1,000 random unit directions reach only 0.45–0.49× of $\sigma_1(S_c)$,

TABLE II: First-order envelope ratio (actual / first-order-predicted equilibrium shift) with increasing ε (10 seeds, S_c-optimal constrained perturbation). Tight at ε=0.01 (≈ 1.00); under-predicts by up to 39% at ε=0.20.

Dataset	ε=.01	ε=.05	ε=.10	ε=.20
Cora	$1.01_{\pm.00}$	$1.07_{\pm.01}$	$1.15_{\pm.03}$	$1.36_{\pm.08}$
Citeseer	$1.01_{\pm.00}$	$1.07_{\pm.01}$	$1.16_{\pm.03}$	$1.39_{\pm.07}$
Pubmed	$1.01_{\pm.00}$	$1.07_{\pm.01}$	$1.15_{\pm.02}$	$1.38_{\pm.06}$
WikiCS	$1.00_{\pm.01}$	$1.01_{\pm.01}$	$1.02_{\pm.01}$	$1.05_{\pm.01}$
Amazon	$1.00_{\pm.00}$	$1.01_{\pm.00}$	$1.03_{\pm.01}$	$1.07_{\pm.01}$

TABLE III: Equilibrium damage at ε=0.10 across attack methods (IGNN, 50-node subgraph, 10 seeds). SVD: AEGIS. Cls-PGD: classification-loss PGD. Shift-PGD: IFT-gradient PGD (solver validation, not an independent baseline).

Dataset	SVD	Cls-PGD	Shift-PGD	Random
Cora	$3.70_{\pm.79}$	$2.51_{\pm.48}$	$3.05_{\pm.70}$	$0.97_{\pm.25}$
Citeseer	$4.63_{\pm.71}$	$2.97_{\pm.42}$	$3.35_{\pm.57}$	$1.01_{\pm.15}$
WikiCS	$2.10_{\pm.22}$	$0.67_{\pm.11}$	$1.93_{\pm.20}$	$0.53_{\pm.09}$

confirming the SVD direction is well separated; an equilibrium-shift PGD attacker [39] with AEGIS's IFT gradients (50 steps) recovers only 72–92% of the closed-form damage; and a classification-loss PGD through the unrolled iteration inflicts 15–70% less equilibrium damage while flipping predictions comparably (0–1.8%), so the SVD direction flips predictions effectively even when optimising a proxy. We therefore describe the SVD direction as the *maximally sensitive perturbation direction* rather than "the optimal attack".

Classification-gradient edges differ. Per-edge classification gradients anti-correlate with discrete ground truth on Cora ($\tau = -0.20_{\pm 0.05}$ vs. AEGIS's $+0.32_{\pm 0.04}$) and stay near zero elsewhere; AEGIS achieves 2–6× higher P@10, confirming that equilibrium and classification sensitivities are distinct vulnerability surfaces.

Discrete edge removal. See fig. 3 for four sequential orderings.

Per-edge finite differences reproduce the S_c column-norm ranking (τ=0.999); recomputing S_c iteratively after each removal closes $\leq$ 3 pp of the static-vs-greedy gap on Cora and $\leq$1 pp on Citeseer/Pubmed at $k\times$ compute, so we keep the static ranking as the headline.

Breach rates. Figure 4 reports prediction-flip fractions across ε. Citeseer and WikiCS stay below 2% across the full range; Cora reaches 7.6% at ε=0.20; Pubmed is the right-skewed outlier (27.4±21.1% mean, median 7.8% vs. mean 10.3% at ε=0.10). Every breached node satisfies $\varepsilon > r_v$, consistent with r_v as a first-order diagnostic rather than a probabilistic certificate (theorem 5).

C. Comparison with Prior Structural Baselines

Table IV reports head-to-head numbers against GR-BCD [5] (PR-BCD is the same family's larger-budget variant), AGNNCert [10], and LODF [38]. AEGIS is a label-free closed-form proxy; the question is how much of each gold-standard signal it retains.

TABLE IV: AEGIS vs. external baselines (10 seeds). GR-BCD [5] is the BCD-family scalable structural attacker (PR-BCD is the same paper's larger-budget variant, see text); AGNNCert [10] is a sound IBP certifier (semantics differ from AEGIS r_v); LODF [38] is the industry DC-PF screener.

Baseline	Setting	AEGIS	Baseline	τ
GR-BCD [5]	Pubmed k=10	0.356	0.350	**+0.69**
GR-BCD [5]	Cora k=5	0.643	1.207	+0.16
AGNNCert† [10]	Cora med. r_v	0.187	1.414	+0.08
LODF [38]	case118 P@10	0.81	$\leq$ 0.20	+0.14

†AGNNCert is a *sound* IBP certificate; AEGIS r_v is a first-order sensitivity threshold (theorem 5). The numerical gap reflects different mathematical objects, not framework dominance.

GR-BCD [5] converges with S_c on Pubmed (τ=+0.69) and diverges on Cora (τ=+0.16), the dataset-level τ acting as a self-diagnostic; PR-BCD [5] (same family, OGB-scale budget) sits beyond our matrix-free boundary. AGNNCert [10]'s IBP radii are 4.9–10.2× looser numerically but mathematically stronger (sound certificate vs. first-order threshold). LODF [38] retargeted to thermal-overload peaks at P@10=0.60 on case57 and collapses on case118; Ejebe–Wollenberg PI [40] is positively correlated but loses to AEGIS on both grids.

D. Phase Transition and Scalability

Phase transition. Figure 6 sweeps $\kappa_{\max}\in[0.30, 0.99]$ on Cora (10 seeds each): the theoretical $\varepsilon_{\text{crit}}$ drops 230× but $\rho(J_z)$ saturates at ≈0.42 (the ReLU pattern blocks the Jacobian from its theoretical maximum), so the resolvent rises only 1.17→1.80 instead of the ∼70× that $1/(1 - 0.99)$ predicts. The phase transition is therefore a theoretical safety boundary, not an empirically observable discontinuity; training converges throughout the sweep, though the post-hoc AEGIS analysis exceeds the 50-step fixed-point budget at $\kappa_{\max}\geq 0.85$.

Scalability and matrix-free self-consistency. Figure 5 compares dense and matrix-free pipelines. The matrix-free $S_c v$ implements the dense S_c up to Neumann truncation; σ_1 agrees within 0.03% at N=200, per-edge τ=0.999, and the analytical truncation residual $\kappa^{200} \in [10^{-105}, 10^{-48}]$ across the suite. For larger N the rSVD error [31] is bounded by the spectral gap; Pubmed (N=19,717) exceeds 24 GB.

E. Ablations and Defense

Subgraph size. Varying $N\in\{30, 50, 100, 200\}$ on Cora (10 seeds): tightness is stable at 1.01–1.02 for $N\leq 100$, degrades to 1.031 at N=200, coverage stays 82–89%. We use N=50 as default (66× faster than N=200, tightness 1.013±0.003). Subgraph-vs-full-graph agreement is high on dense grids (case14 τ=0.80–0.86 at N_{sub}=8–12; case30 τ=0.78±0.07) but low on citation-scale graphs (Cora τ=0.16±0.13 since a 50-node BFS covers ∼1.8% of edges), so citation-scale users should prefer the full-graph pipeline. BFS-centre choice barely matters (τ=0.95±0.02 across highest-vs-median-degree). On the full citation graphs (matrix-free), the AtkAdv ratio over

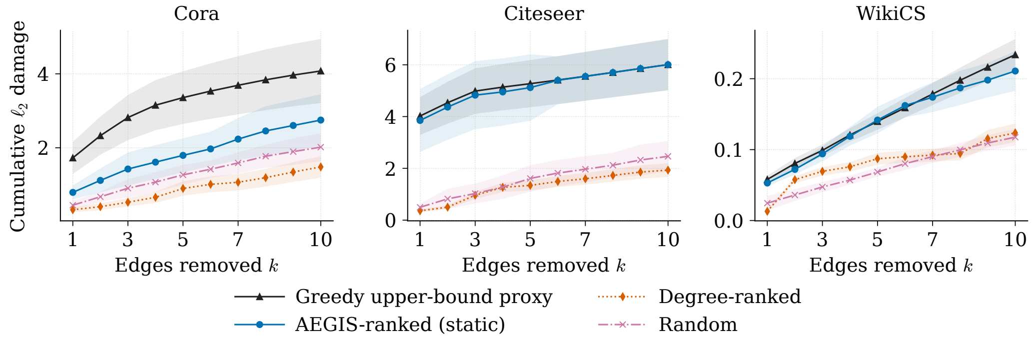


Fig. 3: Cumulative damage from sequential discrete edge removal vs. k (IGNN, 50-node subgraph, 10 seeds, ± 1 sd). AEGIS matches Greedy on Citeseer/WikiCS and reaches 54–67% of Greedy on Cora at k=5–10; degree-ranked stays below random on Cora at every k.

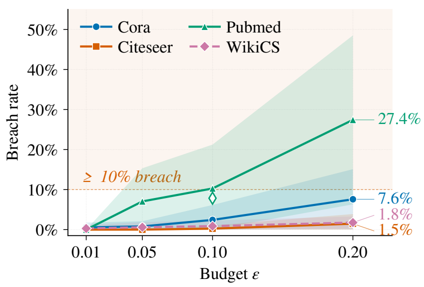


Fig. 4: Breach rate (% of nodes flipped under S_c-optimal perturbation) vs. ε (10 seeds, mean ± 1 sd bands).

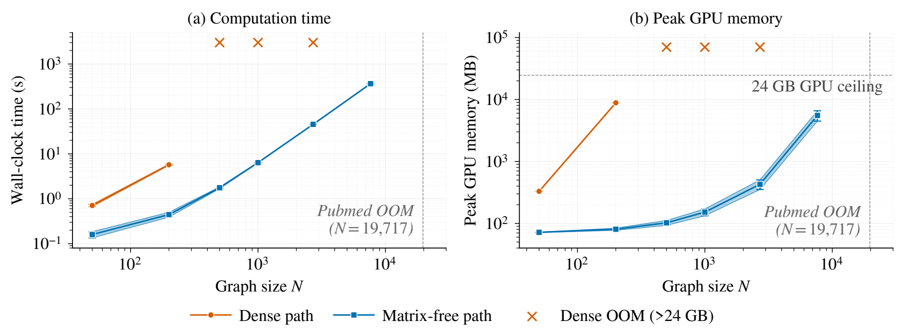


Fig. 5: Dense vs. matrix-free AEGIS pipeline (RTX 4090). Dense scales as $O((Nd)^3)$, exceeding 24 GB beyond N=200; matrix-free is roughly $O(N \log N)$ in time, sub-linear in memory, reaching Amazon Photo (N=7,650) at 365 s/5.5 GB.

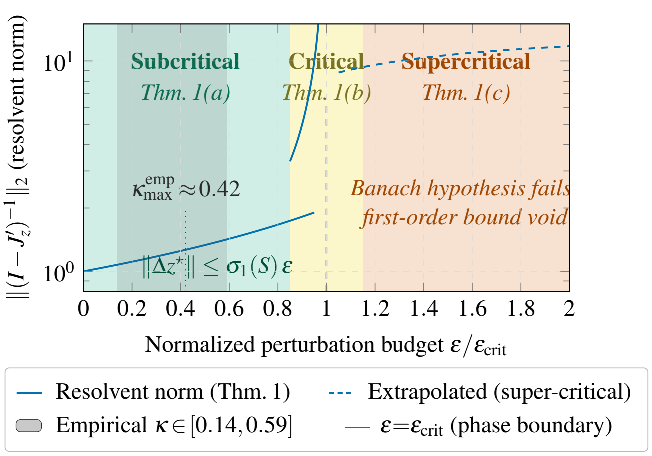


Fig. 6: Three-regime characterisation of theorem 1. The empirical κ band (grey) saturates well below the spectral-normalisation ceiling, so the empirical resolvent growth is far milder than the worst-case $1/(1-\kappa)$ divergence at $\varepsilon \to \varepsilon_{\text{crit}}$.

degree-proportional *grows* to 3.25× (Cora) and 9.82× (Citeseer) at k=10 vs. ≈1.1× on 50-node subgraphs, so the BFS regime is conservative rather than artefactual.

Hyperparameters. Tightness is stable across $d \in \{16, 32, 64, 128\}$; the spectral-norm ceiling $c \in \{0.5, 0.9\}$ traces the accuracy–robustness frontier (r_v=0.090@72.1% vs. 0.051@80.6%).

Defense-informed edge protection (non-adaptive). Masking the top-k edges by v_{ij} (Cora IGNN, N=50, 10 seeds) cuts SVD damage by 42±8% at k=5 and 61±7% at k=10, vs. 11±6%/18±9% for random masking (paired Wilcoxon p<0.002); the attacker is held fixed, so an adaptive variant recomputing S_c would partially erode the gain.

F. Extension to Explicit GNNs

For K-layer explicit GNNs we form $S_K = \partial \text{vec}(Z_K)/\partial \text{vec}(A)$ via finite differences and construct S_c from S_K identically (theorem 7). Table V reports the per-architecture characterisation on Cora: IGNN carries theorem 1; the explicit models receive the computational tool without the regime characterisation.

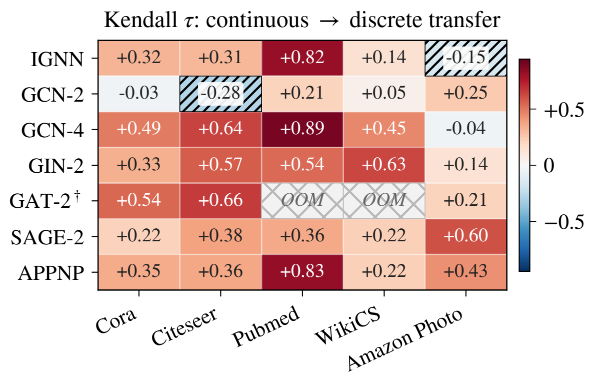
GAT† scope. Standard GAT uses A as a binary mask, so $\partial Z/\partial A_{ij}$=0 on existing edges and S_c is undefined. Our GAT† variant modulates attention by the edge weight,

$$h_i' = \textstyle\sum_{j \in \mathcal{N}(i)} \hat{A}_{ij} \cdot \alpha_{ij} \cdot W h_j, \qquad (12)$$

restoring continuous differentiability with respect to $\hat{A}_{ij}$ and yielding tightness 0.99, AtkAdv 4.4×, τ= + 0.36. Architectures using binary adjacency masks (hard attention, max/min

TABLE V: S_c framework on 7 GNN architectures (Cora, 10 seeds). Tight.: actual/predicted shift ratio. AtkAdv: AEGIS/random damage. τ: Kendall vs. brute-force single-edge-removal truth. Acc: full-graph test accuracy.

Model	Tight.	AtkAdv	τ	Acc
IGNN (eq.)	$1.01_{\pm.00}$	$7.6_{\pm0.5}\times$	$+.32_{\pm.04}$	$77.5_{\pm1.7}\%$
GCN-2	$1.00_{\pm.00}$	$3.6_{\pm0.6}\times$	$-.04_{\pm.03}$	$78.9_{\pm0.9}\%$
GCN-4	$1.02_{\pm.00}$	$3.4_{\pm0.4}\times$	$+.49_{\pm.02}$	$78.3_{\pm1.8}\%$
GIN-2	$1.01_{\pm.00}$	$2.6_{\pm0.1}\times$	$+.33_{\pm.03}$	$76.6_{\pm1.9}\%$
GAT†	$1.01_{\pm.01}$	$2.1_{\pm0.1}\times$	$+\mathbf{.54}_{\pm.06}$	$77.8_{\pm1.2}\%$
SAGE-2	$1.00_{\pm.00}$	$1.9_{\pm0.2}\times$	$+.22_{\pm.10}$	$77.5_{\pm1.5}\%$
APPNP	$1.02_{\pm.00}$	$3.4_{\pm0.3}\times$	$+.35_{\pm.01}$	$\mathbf{82.2}_{\pm0.6}\%$



† *GAT-2: 2-layer attention; OOM on Pubmed and WikiCS.*

Fig. 7: Continuous-to-discrete transfer τ across 7 architectures and 5 datasets (10 seeds, 330 runs). Cell values are mean Kendall τ; buckets: strong (τ>0.5), moderate (0.3–0.5), weak (0–0.3), anti-correlated (τ<0, diagonal hatch); cross-hatch: GPU OOM. Deeper models (GCN-4, APPNP) and GAT† dominate.

aggregation) and GATv2 [41]'s dynamic-attention masking fall outside the framework; the architecture-agnostic property applies only to continuous edge-weight-modulated message passing.

Robust backbones. Under the same $\sigma_1(W)\leq 0.9$ cap, RobustGCN-lite [12] attains $\tau{=}+0.37/+0.54$ (Cora/Citeseer) and GNNGuard-lite [13] $+0.10/+0.53$, matching or exceeding IGNN; the cap is a precondition—without it $\kappa{>}1$ voids theorem 1.

Cross-dataset transfer. Figure 7 reports τ over 5 datasets and 7 architectures (330 runs); $29/33$ cells positive (one-sided sign test $p{<}10^{-5}$). Cold cells (GCN-2 on sparse citation graphs, IGNN on dense product graphs at 50 nodes) recover on full-graph matrix-free analysis (N=7,650 shifts τ from -0.14 to $+0.24$). Heterophilic WebKB [42] GCN-4 attains $\tau=+0.21$–$+0.44$.

Theorem 6's sufficient condition holds for 47–62% of edge pairs; GCN-2 fails the bridge because 2-hop aggregates produce near-uniform first-order shifts. Gradient-based explainers [43] against the same truth give $\tau=-0.07$ to -0.21; the theorem 7 bound is 1.4–$5.9\times$ loose, tighter for shallower models.

VI. Case Study: Power Flow Contingency

As a proof-of-concept cross-domain demonstration (not an operator-grade screening tool), we apply S_c to AC power flow. The correspondence is empirical, not a physical derivation: PTDF/LODF [38] are functions of the DC susceptance matrix B, while v_{ij} is derived from the GNN's learned IFT-resolvent $(I-J_z)^{-1}$. The structural form mirrors DC line-outage factors, but the GNN's J_A is not B^{-1}; the observed correlation reflects the fidelity of the GNN's learned physics. Recent GNN-PF work [44], [45], [46], [3] optimises a fidelity objective; we take the vulnerability-analysis view.

Setup. ContractiveGCN-PF is an IGNN-class operator $F(Z,A){=}\phi(\hat{A}ZW^\top + X_{\text{proj}})$ with spectral-norm-constrained W, trained on five IEEE test cases [36] with a 2-head readout $(\Delta|V|, \Delta\theta)$. Training data are 2,000 load samples per case (70–130% of nominal, *uniform load scaling only*, a narrow envelope that does not cover seasonal peaks, dispatch shifts, or renewable ramps) from PandaPower's [37] AC Newton-Raphson solver; inputs are binary adjacency and 5 bus features (slack/PV indicators, P, Q, $|V|$). Ground truth: ℓ_2 voltage-angle deviation per AC contingency. For case14–118 we use full-graph dense analysis; for case300 a 200-node BFS subgraph. LODF is benchmarked off-spec on this AC voltage-angle truth (its native metric is DC line-flow change), so the comparison is informative on rank order but disadvantages LODF by construction.

table VI reports the envelope ratio and Kendall τ between S_c rankings and brute-force N-1; fig. 8 visualises case14. The envelope is tight in the PF domain ($\approx$ 1.00), and on the credible range (case14–118) ranking correlation is $\tau=+0.37$ to $+0.62$ with P@10=0.66–0.81, identifying 7–8 of the top-10 critical lines. Case300 stress-tests the *GNN's* learning capacity, not S_c scalability: at θ RMSE 0.394 p.u. ($\approx 22.6°$) the surrogate has not learned AC physics; its apparent $\tau=+0.72$, P@10=0.87 are conditioned on an unconverged model and excluded from headline numbers. The matrix-free S_c pipeline itself scales to N=7,650 (fig. 5); improved PF surrogates would extend the demonstration into the operational range.

Baselines and stability. LODF (industry DC screening) attains τ=0.44–0.58 on credible cases, behind AEGIS's 0.62–0.67 on case57/118 (Wilcoxon $p{<}0.01$); CIs overlap on case14/30. The runtime trade-off is unfavorable for operations at this scale (LODF <0.13 s; brute-force N-1 0.1–2 s; AEGIS 2–23 s with training), and operators have impedance data, so the demonstration's value is conceptual: a label-free GNN proxy recovers the dominant N-1 severity pattern from one closed-form SVD pass over a learned representation. v_1 captures one direction, so it surfaces multi-edge vulnerabilities only partially (top-5 overlap 40–64% with N-2 critical sets edge-wise, 7–18% pair-wise). Pairwise rank stability is +0.40–+0.78, higher on larger grids. *Binary adjacency beats admittance-weighted* (P@10=0.81 vs. 0.27 on case118; Spearman between admittance magnitude and N-1 severity is near zero, -0.01 to $+0.15$): a single-line trip removes the line regardless of impedance, so binary sensitivity correctly models the all-or-

TABLE VI: AEGIS on IEEE power-flow benchmarks (10 seeds). Model quality: per-unit RMSE for $|V|$ and angle θ (std $<$ 0.003). τ: Kendall correlation against brute-force N-1; P@10: top-10 precision. Case14–118 is the credible reported range; case300 is a scalability stress-test (model un-converged, not an operational point).

Case	N	$\|E\|$	Model (p.u.) $\|V\|$	θ	AEGIS τ	P@10
Credible operating range						
case14	14	20	.007	.020	$+.42_{\pm.19}$	$.74_{\pm.12}$
case30	30	41	.014	.024	$+.37_{\pm.17}$	$.68_{\pm.06}$
case57	57	78	.033	.059	$+.67_{\pm.09}$	$.66_{\pm.13}$
case118	118	179	.014	.076	$+.62_{\pm.11}$	$.81_{\pm.10}$
Scalability stress-test (model un-converged)						
case300‡	300	411	.031	**.394**⋆	$+.72_{\pm.03}$	$.87_{\pm.06}$

‡200-node BFS subgraph; 1,000 training samples.

⋆θ RMSE $\approx 22.6°$: not a valid operational data point.

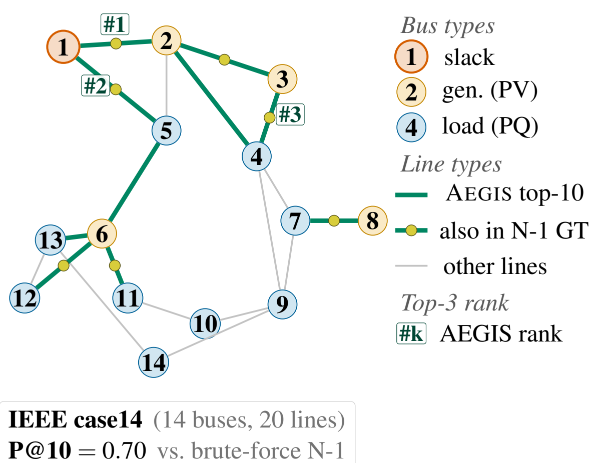


Fig. 8: AEGIS vulnerability ranking on the IEEE 14-bus benchmark (10-seed mean; P@10 $=$ 0.70 for this run, consistent with the 0.74 ± 0.12 in table VI). Bus markers encode physical bus type (slack / generator / load); green edges are the top-10 by v_{ij} with rank tags (#1–#3); yellow midpoint dots mark edges that also appear in the brute-force N-1 top-10 (7 of 10). Top-ranked edges concentrate on the high-flow inter-area tie lines (1–2, 1–5, 7–8), recovering the single-contingency severity pattern from one closed-form SVD pass.

nothing character of N-1. This does not mean impedance is unimportant; sophisticated edge-feature GNNs with impedance, thermal rating, and voltage level as edge attributes are the right next step. Operator-grade deployment requires the extensions noted in the conclusion.

VII. Related Work

We position AEGIS against four threads; each supplies at most one of the three closed-form diagnostics AEGIS produces over the vulnerability spectrum.

Structural attacks. Nettack [1], Mettack [4], GR-BCD/PR-BCD [5], and variants [47] optimise surrogate gradients to flip predictions but provide no closed-form globally maximally sensitive direction. Sober-look studies [15], [16] flag evaluation pitfalls; the defence-informed-masking ablation (section V-E) uses paired Wilcoxon against random masking for this reason.

Certified robustness and defense. Convex relaxations [11], smoothing [6], [7], [8], collective certificates [9], and IBP [10] certify per-node robustness but do not identify the vulnerability-driving edges. Empirical defenses [12], [14], [13] harden models without surfacing a vulnerability spectrum. r_v is a first-order threshold, not a sound certificate (theorem 5); the two are complementary.

Implicit networks and influence functions. DEQs [17], IGNN [19], and well-posedness analyses [26] address input sensitivity $\partial z^*/\partial x$; AEGIS targets structural sensitivity $\partial Z/\partial A$ via the resolvent $(I - J_z)^{-1} J_A$ (implicit; theorem 1) and the unrolled Jacobian S_K (explicit; theorem 7). The closed-form derivative descends from influence-function implicit differentiation [25], extended to adjacency under a constrained projection; GNN explainers [43] target prediction fidelity rather than adversarial vulnerability, and we draw on classical matrix perturbation theory [29], [24].

VIII. Conclusion

AEGIS produces three adversarial diagnostics across the GNN vulnerability spectrum (per-edge rankings, the SVD-optimal attack direction, per-node first-order sensitivity radii) from one closed-form object, the constrained sensitivity matrix S_c, computed matrix-free for any GNN with continuous edge-weight-modulated message passing and supplemented by regime guarantees (theorem 1) for contractive implicit models.

Limitations. (i) Radii r_v are first-order sensitivity thresholds, not probabilistic certificates (theorem 5). (ii) Standard GAT and binary-adjacency-mask architectures fall outside the framework; the seventh architecture is the edge-weighted GAT† variant (section V-F). (iii) On large citation networks 50-node subgraph τ to the full graph is low (τ=0.16 on Cora), but the AtkAdv advantage *amplifies* on the full graph (section V-E); the matrix-free pipeline is the recommended graph-level path. (iv) Spectral-norm-constrained IGNN (77.5% Cora) lags explicit GNNs ($\sim$ 82%); the formal track is a model-class trade-off, not a universal cost. (v) Power-flow τ=+0.37 to +0.62 is proof-of-concept; operator-grade deployment requires admittance-weighted edges, broader operating coverage, and thermal/voltage-violation severity. (vi) The threat model is edge-deletion / weight-perturbation; insertion attacks (Nettack-style) need a candidate-edge basis $\bar{E}$ extension. The vulnerability spectrum informs defence design (section V-E); code will be released with a 90-day stakeholder notification and tiered-access for attack-generation.

References

[1] D. Zügner, A. Akbarnejad, and S. Günnemann, "Adversarial attacks on neural networks for graph data," in *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2018, pp. 2847–2856.

[2] H. Dai, H. Li, T. Tian, X. Huang, L. Wang, J. Zhu, and L. Song, "Adversarial attack on graph structured data," in *International Conference on Machine Learning*, 2018, pp. 1115–1124.

[3] A. M. Nakiganda, C. Weeraddana, N. Popli, and T. van der Hagen, "Graph neural networks for fast contingency analysis of power systems," *arXiv preprint arXiv:2310.04213*, 2023.

[4] D. Zügner and S. Günnemann, "Adversarial attacks on graph neural networks via meta learning," in *International Conference on Learning Representations*, 2019.

[5] S. Geisler, T. Schmidt, H. Sirin, D. Zügner, A. Bojchevski, and S. Günnemann, "Robustness of graph neural networks at scale," in *Advances in Neural Information Processing Systems*, vol. 34, 2021, pp. 7637–7649.

[6] A. Bojchevski, J. Klicpera, and S. Günnemann, "Efficient robustness certificates for discrete data: Sparsity-aware randomized smoothing for graphs, images and more," in *International Conference on Machine Learning*, 2020, pp. 1003–1013.

[7] A. Bojchevski and S. Günnemann, "Certifiable robustness to graph perturbations," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[8] J. Schuchardt, Y. Scholten, A. Bojchevski, and S. Günnemann, "Localized randomized smoothing for GNNs," in *International Conference on Machine Learning*, 2023.

[9] J. Schuchardt, A. Bojchevski, J. Klicpera, and S. Günnemann, "Collective robustness certificates: Exploiting interdependence in graph neural networks," in *International Conference on Learning Representations*, 2021.

[10] Y. Li *et al.*, "AGNNcert: Deterministic certification for graph neural networks against adversarial perturbations," in *International Conference on Learning Representations*, 2025.

[11] D. Zügner and S. Günnemann, "Certifiable robustness and robust training for graph convolutional networks," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2019, pp. 246–256.

[12] D. Zhu, Z. Zhang, P. Cui, and W. Zhu, "Robust graph convolutional networks against adversarial attacks," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2019, pp. 1399–1407.

[13] X. Zhang and M. Zitnik, "GNNGuard: Defending graph neural networks against adversarial attacks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9263–9275.

[14] W. Jin, Y. Ma, X. Liu, X. Tang, S. Wang, and J. Tang, "Graph structure learning for robust graph neural networks," in *Proceedings of the 26th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, 2020, pp. 66–74.

[15] F. Mujkanovic, S. Geisler, S. Günnemann, and A. Bojchevski, "Are defenses for graph neural networks robust?" in *Advances in Neural Information Processing Systems*, vol. 35, 2022, pp. 8940–8952.

[16] L. Gosch, S. Geisler, D. Sturm, and S. Günnemann, "Adversarial training for graph neural networks: Pitfalls, solutions, and new directions," in *Advances in Neural Information Processing Systems*, vol. 36, 2023.

[17] S. Bai, J. Z. Kolter, and V. Koltun, "Deep equilibrium models," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[18] S. Bai, V. Koltun, and J. Z. Kolter, "Multiscale deep equilibrium models," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 5238–5250.

[19] F. Gu, H. Chang, W. Zhu, S. Sojoudi, and L. El Ghaoui, "Implicit graph neural networks," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 11 984–11 995.

[20] T. Miyato, T. Kataoka, M. Koyama, and Y. Yoshida, "Spectral normalization for generative adversarial networks," in *International Conference on Learning Representations*, 2018.

[21] S. Gould, R. Hartley, and D. Campbell, "Deep declarative networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 8, pp. 3988–4004, 2021.

[22] S. W. Fung, H. Heaton, Q. Li, D. McKenzie, S. Osher, and W. Yin, "JFB: Jacobian-free backpropagation for implicit networks," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 6, 2022, pp. 6648–6656.

[23] S. Bai, V. Koltun, and J. Z. Kolter, "Stabilizing equilibrium models by Jacobian regularization," in *International Conference on Machine Learning*, 2021, pp. 554–565.

[24] L. N. Trefethen and M. Embree, *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*. Princeton University Press, 2005.

[25] J. Lorraine, P. Vicol, and D. Duvenaud, "Optimizing millions of hyperparameters by implicit differentiation," in *International Conference on Artificial Intelligence and Statistics*, 2020, pp. 1540–1552.

[26] L. El Ghaoui, F. Gu, B. Travacca, A. Askari, and A. Tsai, "Implicit deep learning," *SIAM Journal on Mathematics of Data Science*, vol. 3, no. 3, pp. 930–958, 2021.

[27] M. Revay, R. Wang, and I. R. Manchester, "Lipschitz bounded equilibrium networks," in *arXiv preprint arXiv:2010.01732*, 2020.

[28] J. Bolte and E. Pauwels, "Conservative set valued fields, automatic differentiation, stochastic gradient descent and deep learning," *Mathematical Programming*, vol. 188, pp. 19–51, 2021.

[29] G. W. Stewart and J.-g. Sun, *Matrix Perturbation Theory*. Academic Press, 1990.

[30] J. R. Magnus and H. Neudecker, *Matrix Differential Calculus with Applications in Statistics and Econometrics*, 3rd ed. Wiley, 2019.

[31] N. Halko, P.-G. Martinsson, and J. A. Tropp, "Finding structure with randomness: Probabilistic algorithms for constructing approximate matrix decompositions," *SIAM Review*, vol. 53, no. 2, pp. 217–288, 2011.

[32] A. Paszke, S. Gross, F. Massa, A. Lerner, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[33] P. Sen, G. Namata, M. Bilgic, L. Getoor, B. Galligher, and T. Eliassi-Rad, "Collective classification in network data," *AI Magazine*, vol. 29, no. 3, pp. 93–106, 2008.

[34] O. Shchur, M. Mumme, A. Bojchevski, and S. Günnemann, "Pitfalls of graph neural network evaluation," in *Relational Representation Learning Workshop, NeurIPS*, 2018.

[35] P. Mernyei and M. Caron, "Wiki-CS: A Wikipedia-based benchmark for graph neural networks," in *arXiv preprint arXiv:2007.02901*, 2020.

[36] IEEE PES, "IEEE PES test feeder working group: Power system test cases," *IEEE Power and Energy Society*, 2024, https://electricgrids.engr.tamu.edu.

[37] L. Thurner, A. Scheidler, F. Schäfer, J.-H. Menke, J. Dollichon, F. Meier, S. Meinecke, and M. Braun, "pandapower – an open-source Python tool for convenient modeling, analysis, and optimization of electric power systems," *IEEE Transactions on Power Systems*, vol. 33, no. 6, pp. 6510–6521, 2018.

[38] A. J. Wood, B. F. Wollenberg, and G. B. Sheblé, *Power Generation, Operation, and Control*, 3rd ed. John Wiley & Sons, 2014.

[39] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *International Conference on Learning Representations*, 2018.

[40] G. C. Ejebe and B. F. Wollenberg, "Automatic contingency selection," *IEEE Transactions on Power Apparatus and Systems*, no. 1, pp. 97–109, 1979.

[41] S. Brody, U. Alon, and E. Yahav, "How attentive are graph attention networks?" in *International Conference on Learning Representations*, 2022.

[42] H. Pei, B. Wei, B. Chang, Y. Lei, and B. Yang, "Geom-GCN: Geometric graph convolutional networks," in *International Conference on Learning Representations*, 2020.

[43] R. Ying, D. Bourgeois, J. You, M. Zitnik, and J. Leskovec, "GNNExplainer: Generating explanations for graph neural networks," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[44] B. Donon, B. Donnot, I. Guyon, and A. Marot, "Graph neural solver for power systems," in *International Joint Conference on Neural Networks*, 2019, pp. 1–8.

[45] V. Bolz, J. Rueß, and A. Zell, "Power flow approximation based on graph convolutional networks," in *18th IEEE International Conference on Machine Learning and Applications*, 2019, pp. 1679–1686.

[46] C. Kim, T. Conrad, R. Karim, J. Oelhaf, D. Riebesel, T. Arias-Vergara, A. K. Maier, J. Jäger, and S. Bayer, "Physics-informed GNN for medium-high voltage AC power flow with edge-aware attention and line search correction operator," *arXiv preprint arXiv:2509.22458*, 2025.

[47] H. Wu, C. Wang, Y. Tyshetskiy, A. Docherty, K. Lu, and L. Zhu, "Adversarial examples on graph data: Deep insights into attack and defense," in *International Joint Conference on Artificial Intelligence*, 2019, pp. 4816–4823.